MILITARY SPECIFICATION

SERVO ELECTRONICS: 10548193

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, quality assurance provisions, and packaging for the servo electronics which is used as subassembly 5A1 and 5A2 in the Power Supply PP-4812\VPS-2, a part of the Radar Set AN\vPS-2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index, of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the **extent specified** herein.

SPECIFICATIONS

MILITARY

MIL-F-13926 - **Fire Control Materiel: General Specification Governing the Manufacture and Inspection of**

MIL-P-14232 - **Parts, Equipment and Tools for Army Materiel; Packaging and Packing of**

MIL-I-45607 - **Inspection Equipment, Supply and Maintenance of**

MIL-C-46357 - **Computer, Gun Direction; M18 Manufacturing** Processes and Inspection of

FSC 1285

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander,US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-A-50687 - Antenna Assembly, AS-2049/VPS-2, 10556312

MIL-S-20708 - Synchro, General Specification for

MIL-S-20708/l - Synchro Control Transformer, Type llCT4e

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-810 - Environmental Test Methods

2.1.2 other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

F10548193 - Servo Electronics
F10556312 - Antenna Assembly AS-2049/VPS-2

PACKAGING DATA SHEET

10548193 - Packaging of Servo Electronics

(Copies of specifications, standards, handbooks, drawing, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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2.2.1 Handbooks.

Military

MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment

(Application for copies should be addressed to Superintendent of Documents? U.S. Government Printing Office~ Washington, D.C. 20402)

3. REQUIREMENTS

3.1 **Fabrication. The servo electronics shall be manufactured in accordance with Drawing F10548193 and drawings pertaining thereto.**

3.2 General specifications. Where applicable, the servo electronics shall meet the requirements of MIL-F-13926-and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. **order of precedence**

b. Dimensions and tolerances

c. **Inorganic protective surface finishings**

d. **Part identification and marking**

e. **Workmanship**

f. Soldering electrical connections

3.2.2 MIL-C-46357.

a. **Etched circuit assemblies**

b. Wiring

c. **Threaded fasteners**

d. Marking

e. **Finishes**

3.3 Environmental service conditions.

3.3.1 **Vibration. The servo electronics shall be mounted on the vibration table in normal service position. The servo**

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electronics shall meet the requirements of 3.4 and 3.5 after being subjected to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range of 10 to 40 hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency at each frequency range shall not exceed 1 Hz per second. vibration tests shall be conducted under ambient **room conditions.**

3.3.2 Shock. The servo electronics shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The half-sine shock impulses shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 ± 1 milliseconds (ins). The G value of each impulse shall be within ± 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall **occur at approximately 5.5 ms.**

3.3.3 Altitude.

3.3.3.1 Operating. The servo electronics shall meet e requirements of 3.4 and 3.5 during and after exposure to bar metric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. The servo electronics shall meet the requirements of 3.4 and 3.5 after exposure for not less than one hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The servo electronics shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°C and +71°C.

3.3.4.2 Nonoperating (storage). The servo electronics shall not suffer permanent damage when subjected to not less than four hours at each temperature extreme of -54°c and +80°C, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (nonoperating). The servo electronics shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity of 95 ±5 percent for a period of not less than 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the servo electronics.

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3.5 Performance.

3.5.1 Operating. The servo electronics shall perform within the following specification when operating from a dc power supply voltage from 20 to 30 volts.

3.5.1.1 Spike survival. The unit shall survive a **spike up to 80 volts (ground reference), having a width of 0.5 to 1.0 milliseconds at** a rate of three (3) per minute for a duration of two (2) **minutes, on the input dc power (nominal), and perform normally as specified in requirements** of **3.5.**

3.5.2 Inputs. The servo electronics shall have three inputs which are summed to control the output. The inputs shall be as follows:

Synchro error - Ac v0ltage
Tach signal - Dc voltage
Dc Signal No. 2 - Dc voltage

3.5.3 Input impedance. The input impedance for the inputs shall be as follows:

Tach signal - Not less than 10 k ohms
Synchro error - Not less than 50 k ohms
Dc signal No. 2 - Not less than 50 k ohms

3.5.4 Input overload. The synchro error signal input will be driven by a MIL-S-20708/l synchro and shall not be damaged by the maximum possible output from the CT synchro (approximately 57 volts into an open circuit)~

3.5.5 Isolation. The input circuit shall be dc isolated from the output.

3.5.6 Nu1l voltage. With the synchro error, tach and dc signal No. 2 inputs shorted, the output voltage shall not exceed 500 millivolts under all operating conditions.

3.5.7 Quadrature rejection. The response to a quadrature signal shall be at least 40 dB below the response to an in-phase signal of the same amplitude.

3.5.8 Reference signal. The reference signal will be 115 Vac ± 5%, 400 Hz ± 1%.

3.5.9 Reference isolation. The reference and power supply returns shall be isolated from the signal return.

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3.5.10 Preamplifier - demodulator gain. The synchro error preamplifier shall have a gain of 10.3 Vdc per Vac ± 5% for all output voltages of 3.0 Vdc or less, measured at the DEMOD OUTPUT test point with respect to SIG COM test point. The tolerance shall be ± 10% for the requirements of 3.3.4.1.

3.5.11 Preamplifier - demodulator bandwidth. The gain at 80HZ shall be no more than 3 db down from the DC gain. . The phase lag of the output, with respect to the input, shall not be more than 450 at 80 Hz. Measurements should be made by using a suppressed carrier signal source such as that available from a Servo Analyzer. The demodulator should be loaded by the Compensation Circuitry during this test.

3.5.11.1 Compensation circuitry. The two wire demodulator out-put signal shall be applied to the compensation network as shown in Figure 3. The signal may be further attenuated by activation of a small signal miniature relay. The second pole of this relay shall be used to reduce current limit level as specified in 3.5.18.

3.5.11.2 phase sensitivity. The servo electronics shall be phase sensitive to the synchro error signal with respect to the reference signal. The synchro error signal being in-phase with the reference signal shall produce a positive output voltage. with an 180° out-of-phase synchro error signal input, the output shall be a negative output voltage.

3.5.12 Phase sensitivity. The servo electronics shall be phase sensitive to the synchro error signal with respect to the reference signal. The synchro error signal being in-phase with the reference signal shall produce a positive output voltage. with an 1800 out-of-phase synchro error signal input, the output shall be a negative output voltage.

3.5.13 Amplifier gain. The gain for each summing input shall be as follows:

Synchro channel - Demodulator gain 10.3 volts/volt ± 5%
Main amplifier gain 47 volts/volt ± 5%

Tach channel - Main amplifier gain + 1.68 volts/volt ± 5%

Dc signal No. 2 - 16.6 volts/volt ± 10%

The gain tolerance shall be ± 10% for the requirements of Paragraph 3.3.4.1. The output voltage at J2-K with respect to J2-J shall be positive for -

1) Positive signal at J1-K with respect to J1-U.

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2) Positive signal at J1-M with respect to J1-L.

3) Positive signal at DEMOD OUTPUT with respect to SIG COM.

The output voltage at J2-K with respect to J2-J shall be negative for -

1) Negative signal at J1-K with respect to J1-U.

2) Negative signal at J1-M with respect to J1-L.

3) Negative signal at DEMOD OUTPUT with respect to SIG COM.

3.5.14 Gain stability. The overall gain stability, as a function of the input voltage, shall be ± 10%. The tolerance shall be ± 20% for the requirements of 3.3.4.1.

3.5.15 Linearity. The overall voltage gain linearity shall be 10%. The tolerance shall be ± 20% for the requirements of 3.3.4.1.

3.5.16 Limiting. The preamplifier-demodulator output voltage shall limit at 9 ± l Vdc when the unit is overdriven up to the maximum input as specified in 3.5.40

3.5.17 Dynamic performance. The servo electronics shall be connected to an Antenna Assembly AS-2049/VPS-2~ 10556312, and shall meet the requirements of 3.5.6 of MIL-A-50687 (MU) as part of the requirements for 4.2. On a regular production basis, a qualified manufacturer shall demonstrate that the servo-amplifier is capable of driving a motor-tachometer combination similar to the Procurement Activity, P/N 10548163.

3.5.18 Output power and limiting. The servo electronics shall be capable of the following with a resistive load of one ohm's.

1) At least 14 Vdc output at 4.5A continuous under all specified operating conditions.

2) At least 15 amps output continuously for ambient temperatures below -18°C.

3) At least 15 amps output surge for one second non-repetitive for ambient temperature above -18°C.

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The servo electronics shall contain internal current limiting as follows:

14.() ± 1.75 A at 71°C.

15.0 ± 1.75 A at 20°C

16.0 ±.75 A.at -40°C

3.5.19 Limiting mode relay switching. Grounding the limit switch line to power ground shall energize a relay and reduce the current limit level to the following:

4.0 ±.5 A at 71°C

4.25 ±.5 A at 20°C

4.5 ± .5 A at -40°C

The response time shall be less than 10 ms from the condition of 3.5.18.

3.5.20 Ripple. With a dc output of plus or minus 5 Vdc, the peak-to-peak ripple shall not exceed 0.5 Vat.

3.5.21 Stow mode relay switching. Grounding the stow control line to power ground shall energize a relay and the DEMOD OUTPUT voltage shall be + 3.1 ± 20% Vdc with respect to SIG COM.

3.5.22 Up-down relay switching. With the condition of 3.5.21 grounding, the up/down control line to power ground shall energize a relay and the DEMOD OUTPUT voltage shall be -3.1 ± 20% Vdc with respect to SIG COM.

3.5.23 Test points. The servo electronics shall have test points mounted as shown in Drawing F10548193, sheet 4. The test points shall be connected so as to monitor (measure) the following signals:

1) SIG COM - synchro error, tach and demodulator signals.

2) SYNC SIG - Synchro error input signal, a 400 Hz ac signal in-phase or l80° out-of-phase with reference 400 Hz ac signal of normally less than 1.0 Vac.

3) DEMOD OUTPUT - Demodulator output signal, a positive dc voltage of normally less than 20 Vdc when the synchro error input is in-phase (with reference signal) and a negative dc voltage of normally less than 20 Vdc when the synchro error input is out-of-phase.

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4) TACH SIG - Tachometer input signal, a positive or negative dc voltage of normally less than 1.0 Vdc.

5) OUTPUT HI - Motor voltage output, a positive or negative dc voltage of normally less than 28 Vdc (with reference to J2-J).

6) OUTPUT LO - Motor voltage output return, the dc return for the OUTPUT HI (J2-K).

With the exception of SIG COM, the test points shall be connected to the respective signals through 10 k ohm isolation resistors.

3.6 Reliability. The servo electronics shall have a mean time between failure (MTBF) of 10,000 hours minimum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three servo electronics manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. when the Government is responsible for conducting first article approval tests, the contractor, prior to submitting-the sample to the-Government, shall inspect the-sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance of materials.

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4.2.2 **Contractor testing.** When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves "the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 **Components and subassemblies.** All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provision (SQAP listed in the technical data package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 and 4.4 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.3.2.3 and 4.4 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 **Functional tests.** The requirements and tests in Table I shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature (24°c ± 8°c).

TABLE I. CLASSIFICATION OF DEFECTS
(100 percent inspection)

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURES
CRITICAL:	**NONE**		
MAJOR			
101	Electrical	3.4	4.6.2
102	Performance	3.5	4.6.3
MINOR:	**NONE**		

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4.3.3 **Acceptance and rejection.** Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. One servo electronics shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table II. Three power supplies shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table III.

TABLE II. SPECIAL SAMPLING

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301 Fabrication	3.1	Applicable drawings - Visual
302 General specifications	3.2	MIL-F-13926 and MIL-C-46357 - Visual

TABLE 111. SPECIAL SAMPLING, ENVIRONMENTAL

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
303 Temperature	3.3.4	4.6.1.4

4.4.1 **Failure of sample.** Should any one item of a special sampling fail to meet the specified test requirements, acceptance ***of*** **the product shall be suspended by Government until necessary corrections have been made** by **the contractor and a resubmitted** sample has been approved (see 4.3.3) .

4.5 **Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the** applicable requirements of MIL-I-45607. The government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.5.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 **Contractor-furnished inspection equipment.**

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4.5.2.1 **Government design.** **Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical** data listed in the **TDP.**

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures) specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment design **must be considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.**

4.5.2.3 Test equipment. In conjunction with 4.5.2.2, the standard test equipment listed in Table IV shall be utilized in **Performing the applicable test of 4.6.**

TABLE IV. TEST EQUIPMENT LIST

ITEM NUMBER		DESCRIPTION
1.	Power supply	0-30 Vdc at 20 A.
2.	**Voltmeter, digital**	**0.1 mVdc to** 100 **Vdc 0.1% ± 1 digit.**
3.	**Voltmeter, digital, ac**	**4-1/2 digits, 200mV range accuracy: ± (0.2% of input +0.16mV) 2V range accuracy: ± (0.2% of input to .3mV)**
4.	**Oscilloscope**	**Input Sensitivity: .05 to 50 V/division. **Horizontal Deflection: 10 us\division. **Vertical Bandpass: Dc to 4** Mhz.

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TABLE IV. TEST EQUIPMENT LIST (Cont'd)

ITEM NUMBER		DESCRIPTION
5.	Multimeter	Resistance Range: xl, XIOO. XIOO.
6.	Test fixture, servo electronics	As specified in Figure 1, see Table V for fixture parts list.
7.	Frequency Response Analyzer	Frequency Range: .OIH$_z$ to 10KH$_z$ Accuracy: 1% of setting Output Signal: Type: Suppressed carrier modulation Amplitude: 10.0 volts peak. Accuracy: 1% Distortion: l%RMS at full scale Frequency Range: 50H$_z$ to 10KH$_z$ Phase Error: 10 to 5KHZ Phase Readout: 3-1/2 digit to ± 180.Oo to nearest O.1° Amplitude Readout: Percent full scale O to 150% to nearest 0.1%. Amplitude Ratio: Resolution: 0. ldB Accuracy: 0.15dB
8.	Pulse generator	Pulse Width: 0.5 to l.oms MIN Pulse Amplitude: 8 0V Transition Time: 15ns Manually triggerable
9.	Variable autotransformer	0-132 Vat, 400 Hz.
10.	Antenna assembly	AS-2049/VPS-2, 10556312
11.	Mount, cabling, and testing equipment	Refer to MIL-A-50687 (AR)
12.	Motor-Tachometer	Part No. 10548163

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TABLE V. SERVO ELECTRONICS TEST FIXTURE PARTS LIST

ITEM	DESCRIPTION
c l	Capacitor, 100 uF, ± 20%, 20VQ
C2	Capacitor, 0.33 ufd, ± 20%, 20V.
CR1	Diode, IN645
CR2-3	Diode, IN746A
F1	Fuse holder and 1 A fuse
P1	Connector, MS3116E14-19S
P2	Connector, MS3116E14-12S
R1-2	Potentiometer, 1 k ohms ± 10%, 2 watt
R3	Resistor, selected (24 k ohms ± 5% 1/4 W nominal
R4	Resistor, 2.7 k ohms ± 5%, 1/4 watt
R5	Resistor, 390 ohms ± 5%, 1/4 watt
R6	Resistor, 1 ohm ± 5%, 100 watt
R7	Resistor, 5 ohm ± 5%, 200 watt
R8	Resistor, 1 k ohms ± 5%, 1/4 watt
R9	Resistor, 33 k ohms ± 5%, 1/4 watt
R10-R1l	Potentiometer, 500 ohms ± 10%, 2 watt
S1	Switch, 4PST - Ckutler/Hammer-7660K13 or equivalent
S2	Switch, 4 position rotary
S3-S7	Switch, SPST 2 A dc
S8-S9	Switch, DPDT 2 A dc
Slo	Switch, DPDT, 20 A de
T1	Transformer, filament, 400 Hz, 115 Vac primary, 7.5/6.3 Vac secondary
TP1-TP8	Pin jack, E.F. Johnson - 105-0800 or equivalent

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4.6 Test methods and procedures.

4.6.1 Environmental. Environmental test shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.6.1.1 Vibration. For conditions specified in 3.3.1 and that failure criteria do not apply after vibration, the servo electronics shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.1.

4.6.1.2 Shock. For conditions specified in 3.3.2. After the shock impulses, the servo electronics shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.2.

4.6.1.3 Altitude (nonoperating and operating). This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in step 2 the altitude shall be limited to 40,000 feet (5.5 inches Hg). operation of step 3 shall be the performance tests specified in 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and nonoperating). This test shall be performed as specified in procedure I, Method 501 of MIL-STD-810. The temperature of step 2 shall be +80°C. The temperature of step 4 shall be +71°C. In step 5, the temperature shall be maintained for four hours and the servo electronics shall be tested in accordance with 4.6.2 and 4.6.3 during this time.

4.6.1.4.2 Low (operating and nonoperating). This test shall be performed as specified in procedure I, Method 502 of MIL-STD-810. the temperature of step 2 shall be -54°c maintained for not less than four hours. The temperature of step 4 shall be -40°C. In steps 5 and 7, the servo electronics shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity (nonoperating) . This test shall be performed in accordance with procedure II, Method 507 of MIL-STD-810. The servo electronics shall be tested in accordance with 4.6.3.

4.6.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4*

4.6.3 Performance. The servo electronics shall be tested using the test set-up as shown in Figure 2, except those tests in 4.6.3.1, 4.6.3.2, and 4.6.3.3 with referenced items of test equipment from Table IV. use 5-ohm load, except in those tests specifying l-ohm load. During testing keep the base plate temperature below 70°C.

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4.6.3.1 Input impedance. Connect item 5, multimeter, between the following pins and verify compliance with 3.5.3:

SIGNAL	INPUT	RETURN
Synchro error	J1-A	J1-B
Tach	J1-M	J1-L
Dc signal No. 2	J1-K	J1-U

4.6.3.2 Reference isolation. Connect item 5, multimeter, between J1-F and verify compliance with 3.5.9.

4.6.3.3 Isolation. Connect item 5, multimeter, between J1-A and J2-K and verify compliance with 3.5.5.

4.6.3.4 Reference signal. Connect item 5, multimeter, between J1-E and J1-F and verify compliance with 3.5.8.

4.6.3.5 Null voltage. Short all inputs to their respective return and verify compliance with 3.5.6, using item 2.

INPUT	PINS SHORTED
Synchro error	J1-A to J1-B
Tach	J1-M to J1-L
Dc signal No. 2	J1-K to J1-U

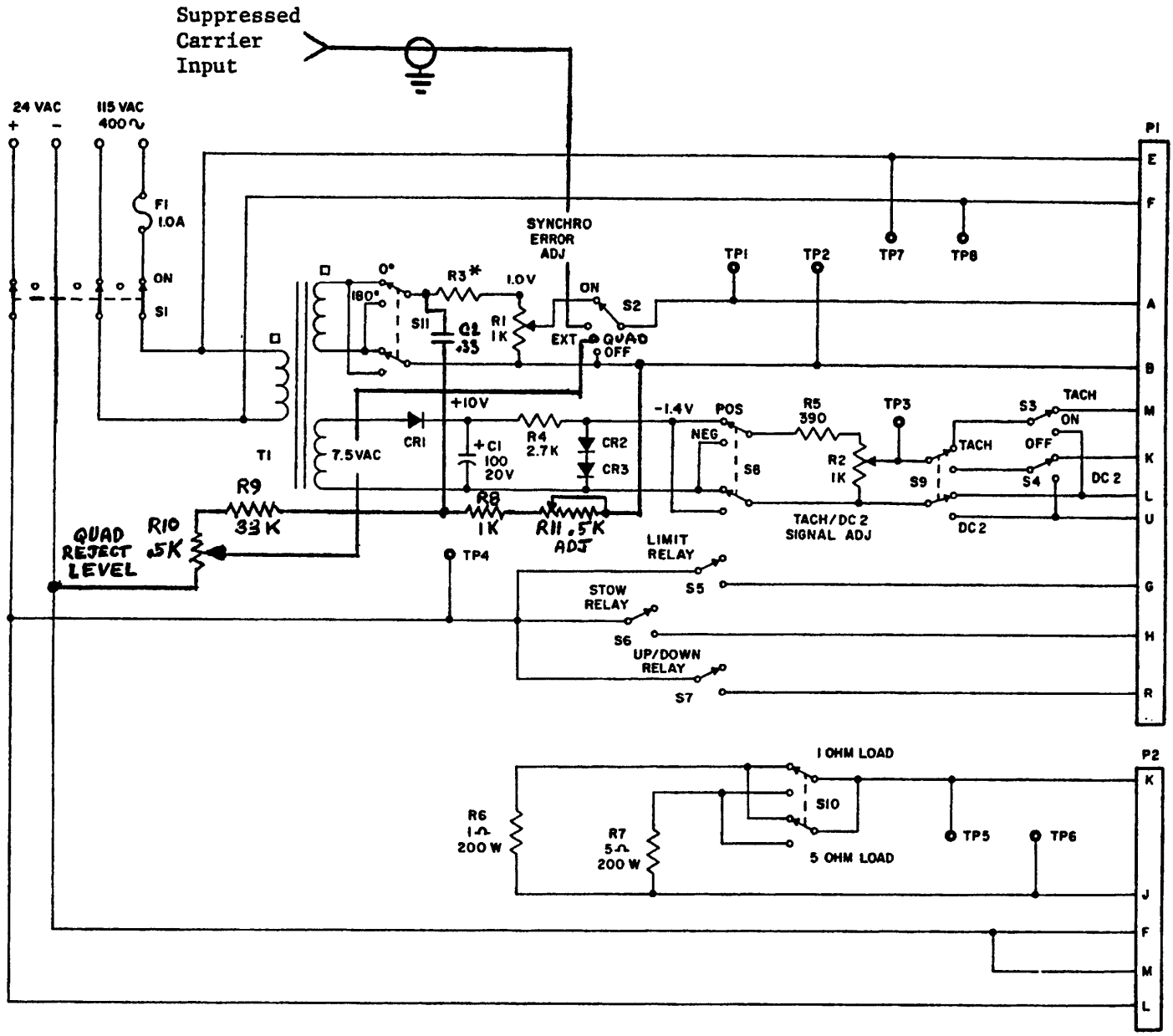
4.6.3.6 Preamplifier-demodulator gain and phasing.

4.6.3.6.1 Connect item 2, voltmeter, to DEMOD OUTPUT and SIG COM test points. Adjust the servo electronics test fixture (refer to figure 1) for an ac signal in-phase with respect to reference signal. Set the amplitude to obtain +3.0 Vdc at the DEMOD OUTPUT with respect to SIG COM test points. Verify gain compliance with 3.5.100

4.6.3.6.2 connect item 2, voltmeter, to J2-K with respect to J2-J and verify polarity compliance with 3.5.13.

4.6.3.6.3 Connect item 2, voltmeter, to DEMOD OUTPUT and-SIG COM test points. Adjust the synchro error to 180° (out-of-phase), with respect to reference ac signal, and set the amplitude to obtain -3.0 Vdc at the DEMOD OUTPUT with respect to SIG COM. Verify gain compliance with 3.5.10.

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*Selected for 1 volt P-P output

Figure 1 Servo Electronics Test Fixture

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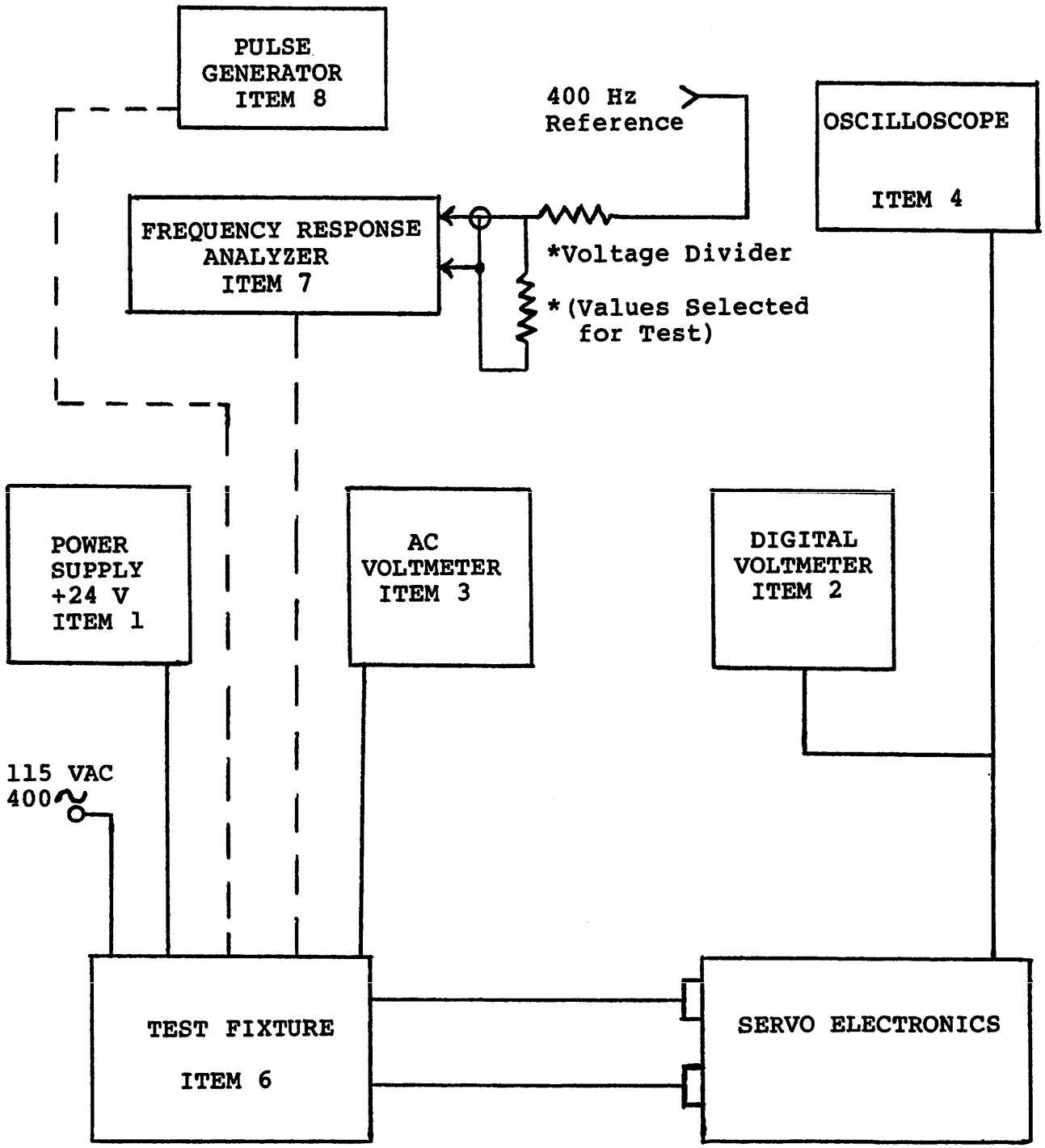
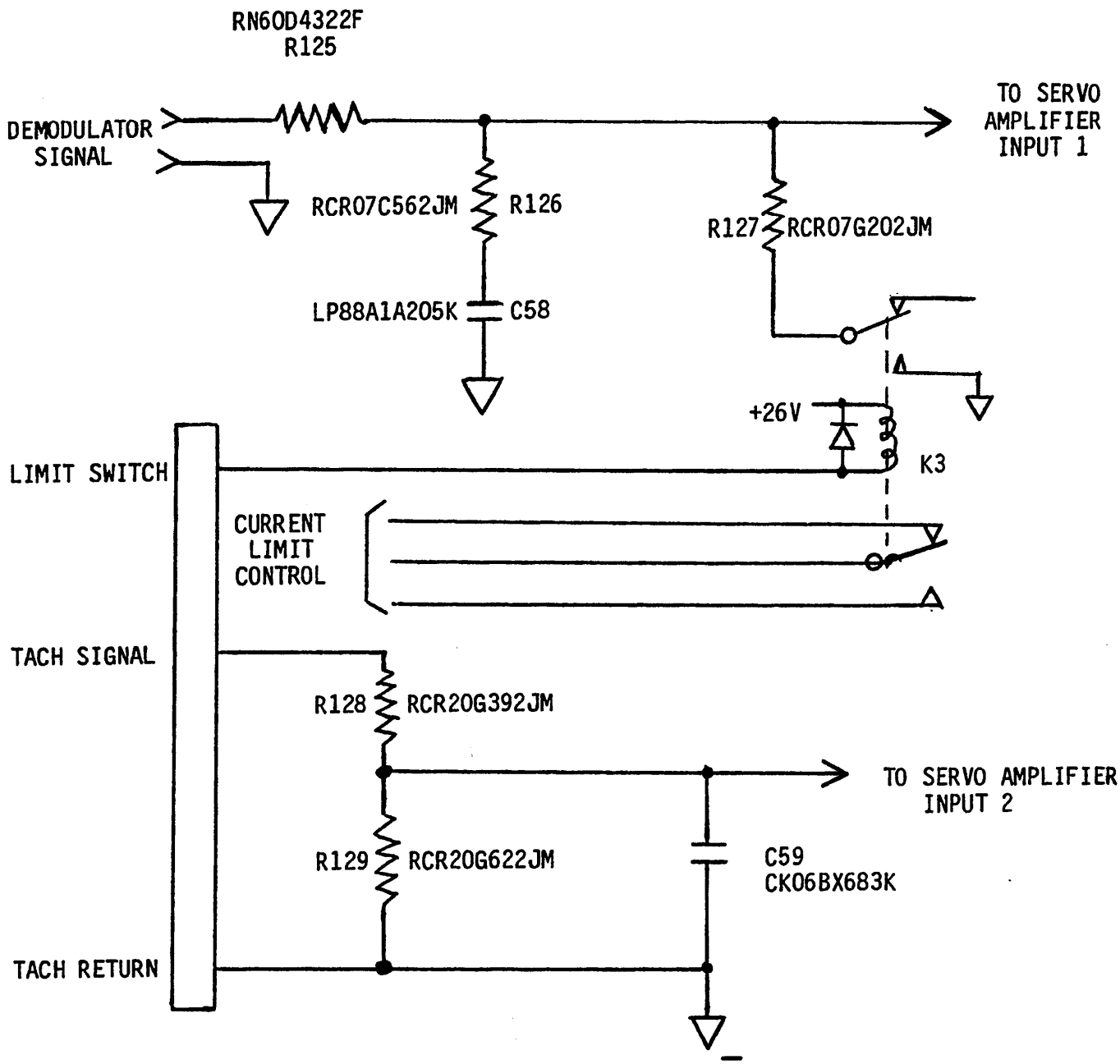


Figure 2. Test set-up, servo electronics

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All type designations, for reference only.

Figure 3. Compensation Circuitry

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4.6.3.6.4 Connect item 2, voltmeter, to J2-K with respect to J2-J and verify polarity compliance with 3.5.13.

4.6.3.7 Gain **and polarity. Any input not being used during the following tests shall be shorted to its return.**

4.6.3.7.1 Connect item 2, voltmeter, between J2-K and J2-J. Adjust synchro error for 0° (in-phase), with respect to reference ac signal across J1-A with respect to J1-B to produce +10.0 Vdc at J2-K with reespect to J2-J. Verify gain compliance with 3.5.13.

4.6.3.7.2 **Increase the synchro error to produce +17.0 Vdc at J2-K with respect to J2-J and verify gain and polarity compliance with 3.5.12 and 3.5.13.**

4.6.3.7.3 Adjust the synchro error for 180° (out-of-phase with reference to ac signal) across J1-A with respect to J1-B and adjust the amplitude to produce -10.0 Vdc at J2-K with respect to J2-J . Verify gain compliance with 3.5.13.

4.6.3.7.4 Increase the synchro error to produce -17.0 Vdc at J2-K with respect to J2-J and verify gain and polarity compliance with 3.5.12 and 3.5.13.

4.6.3.7.5 **Apply a positive dc tack signal voltage to J1-M with respect to J1-L and adjust the amplitude to produce +10.0 Vdc at J2-K with respect to J2-J. Verify gain compliance with 3.5.13.**

4.6.3.7.6 Increase the tach signal input voltage for +17.0 Vdc at J2-K with respect to J2-J and verify gain and polarity compliance with 3.5.13.

4.6.3.7.7 **Adjust the tach signal for a negative dc voltage applied to J1-M with respect to J1-L to produce -10.0 Vdc at J2-K with respect to J2-J. Verify gain compliance to 3.5.13.**

4.6.3.7.8 **Increase the dc tach signal voltage for -17.0 Vdc at J2-K with respect to J2-J and verify gain and polarity compliance with 3.5.13.**

4.6.3.7.9 Apply a positive dc signal No. 2 voltage to J1-K with respect to J1-U to produce +10.0 Vdc at J2-K with respect to J2-J. Verify gain compliance with 3.5.13.

4.6.3.7.10 Increase the dc signal No. 2 input voltage for +17.0 Vdc at J2-K with respect to J2-J and verify gain and polarity compliance with 3.5.13.

4.6.3.7.11 Adjust the **dc signal No.** 2 for a negative dc voltage at J1-K with respect to J1-U to produce -10.0 Vdc at J2-K with respect to J2-J. Verify gain compliance with 3.5.13.

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4.6.3.7.12 Increase the dc signal No. 2 input voltage for -17.0 Vdc at J2-K with respect to J2-J and verify gain and polarity compliance with 3.5.13.

4.6.3.7.13 Performance of tests of 4.6.3.7 verifies compliance to 3.5.2.

4.6.3.8 Linearity. Using the gain measurement recorded in 4.6.3.7, verify compliance with 3.5.15.

4.6.3.9 **Gain** stability. Adjust the ac synchro error input to obtain +10.0 Vac at J2-K with respect to J2-J. Adjust the +24 V power supply (item 1) for +20.0 Vdc and record the output voltage across J2-K and J2-J. Readjust the power supply for +30.0 Vdc and record the output voltage. Verify compliance with 3.5.14.

4.6.3.10 **Spike survival.** Repeat the test of 4.6.3.9 with the following exception:

a. Input voltage at +24Vdc (nominal)

b. Impose a spike of 80°V (from ground reference) having a width of 0.5 to 1.0 millisecond at a rate of 3 per minute prior to performing the test of 4.6.3.9 using Item 8.
Verify compliance with 3.5.1.

4.6.3.11 Input overload and limiting.

4.6.3.11.1 Using item 9, variable transformer, connected to error signal (external) , apply a 400 Hz ac signal of 57 V peak-to-peak *(open* **circuit) across J1-A and J1-B. Connect item 2, voltmeter, to J2-K with respect to J2-J and verify compliance with 3.5.16.**

4.6.3.11.2 **Remove the 400 Hz, 57 Vac signal and perform the test of 4.6.3.7.1 to verify compliance with 3.5.4.**

4.6.3.12 Quadrature rejection.

4.6.3.12.1 Connect item 3 to TP1 and TP2 of the test fixture and adjust the in-phase error signal for a +10.0 Vdc output from the servo amplifier. Measure and record the error signal at TP1 and TP2 and the DEMOD-OUTPUT level.

4.6.3.12.2 With the quadrature signal level equal to, but 900 out of phase with the error signal level required to obtain the +10.OVDC of 4.6.3.12.1, measure and record the DEMOD OUTPUT level with respect to SIG COM. Compliance to 3.5.7 will be confirmed if the DEMO OUTPUT level is equal to or less than one-one-hundredth (1/100) of the DEMOD OUTPUT signal level recorded in 4.6.3.12.1.

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4.6.3.13 Output power and limiting.

4.6.3.13.1 Disconnect the 5-ohm load connected to J2-K and J2-J and connect a l-ohm load. Adjust the +24 V power supply for 20.0 Vdc .

4.6.3.13.2 Adjust the synchro error to 0° (in-phase) for +14.0 Vdc output voltage at J2 with respect to J2-J and verify compliance with 3.5.18. Adjust synchro error to 180° (out-of-phase) for -14.0 Vdc output voltage at J2-K with respect to J2-J and verify compliance with 3.5.18.

4.6.3.13.3 Adjust the tach signal (positive) to J1-M with respect to J1-L for +14.0 Vdc output voltage at J2-K with respect to J2-J and verify compliance to 3.5.18. Adjust the tach signal (negative) for -14.0 Vdc output voltage at J2-K with respect to J2-J and verify compliance with 3.5.18.

4.6.3.13.4 Adjust the dc signal No. 2 (positive) to J1-K with respect to J1-U for +14.0 Vdc output voltage at J2-K with respect to J2-J and verify compliance with 3.5.18. Adjust the dc signal No. 2 (negative) for -14.0 Vdc output voltage at J2-K with respect to J2-J and verify compliance with 3.5.18.

4.6.3.14 (Current limiting mode. With the conditions Of 4.6.3.13.1 and 4.6.3-.13.2, except the power supply set for +24.0 Vdc, connect pin J1-G to J2-L and verify compliance with 3.5.19.

4.6.3.15 Dynamic performance. Disconnect P2 from J2 of the servo electronics and connect the antenna item 10 to J2. Set synchro error switch to EXT and connect the input signal of 3.5.6 of MIL-A-50687 (AR) to the synchro error, signal test points TP1 and T 2. Perform the test as described in 4.6.3.6 of MIL-A-50687 (AR) using the servo electronics to first operate the elevation drive and then the traverse drive to meet the requirements of 3.5.6 of MIL-A-50687 (AR).

4.5.3.16 Output ripple. Connect the 5-ohm load resistor across J2-K and J2-J. Use the synchro error (0° and 180°) to produce a +5.0 Vdc and -5.0 Vdc output at J2-K with respect to J2-J and verify compliance with 3.5.20.

4.6.3.17 Stow mode. Connect pins J1-H to J2-L together and using item 2, voltmeter, connected to DEMOD OUTPUT with respect to SIG COM test points, verify compliance with 3.5.21.

4.6.3.18 Up-down mode. With the existing condition of 4.6.3.17, connect pins J1-R and J2-L together and verify compliance with 3.5.22.

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4.6.3.19 Test **points**.

4.6.3.19.1 Using 0° synchro error, apply 0.5 Vac across J1-A with respect to J1-B as measured with item 3, voltmeter. Connect item 2, voltmeter to SYNC SIG and SIG COM test points and verify compliance with 3.5.23.

4.6.3.19.2 **Using** positive tach signal, apply a 0.5 Vdc across J1-M with respect to J1-L. Connect item 2, voltmeter, to TACH SIG and SIG COM test points and verify compliance with 3.5.23..

4.6.3.19.3 Adjust the tach signal (positive) to produce +10.0 Vdc at J2-K with respect to J2-J. Connect item 2, voltmeter, to OUTPUT HI with-respect to OUTPUT LO test points and verify compliance with 3.5.23.

4.6.3.19.4 **performing the test of 4.6.3.6 will verify the compliance of DEMOD OUTPUT test points with 3.5.23.**

4.6.3.20 Phase sensitivity.

4.6.3.20.1 Adjust the synchro error (positive) for 0.5 Vac in-phase with J1-E (reference) at J1-A with respect to J1-B. Connect item 2, voltmeter, across the output pins, J2-K with respect to J2-J and verify compliance with 3.5.12.

4.6.3.20.2 Apply a synchro error signal of 0.5 Vac out-of-phase with J1-E (reference) at J1-A with respect to J1-B. Check the output voltage pins, J2-K with respect to J2-J for compliance with 3.5.12.

4.6.3.21 Preamplifier - demodulator bandwidth.

4.6.3.21.1 With the 400 hertz reference input and the output and return signal of frequency response analyzer connected together, calibrate the frequency response analyzer amplitude for suppressed carrier operation according to frequency response analyzer manual.

4.6.3.21.2 With Frequency Response Analyzer Output connected to Suppressed Carrier Input on the test fixture and the Frequency Response Analyzer Input connected to DEMOD OUTPUT, adjust the analyzer output signal level for 25 millivolts at a frequency of 20 hertz.

4.6.3.21.3 **Maintaining the 25 mv input level; vary the frequency of the Frequency Response Analyzer from one hertz (1 Hz) through eighty hertz (80 Hz), and verify compliance to 3.5.11.**

4.7 Reliability. A reliability demonstration test shall be conducted as part of FA/lPT, to assure compliance with the requirement of 3.6 at 90 percent confidence level. A minimum of 20 hours

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of operation on each sample (sample size 6 minimum) shall be accumulated after/during exposure to each nonoperating/operating environment of this specification (3.3).

Additional hours are required at 24° ± 8°C in order to accumulate a minimum of 3000 hours on each sample. The sample servo electronics shall be tested until the total hours (23,025 minimum) satisfies the following relationship or until 6 failures have occurred or 92,750 hours (maximum) have been accomplished.

$$MTBF \geq \frac{2T}{X^2 90,\ 2\ (r + 1)}$$

T = Total hours.

$X^2 90,\ 2\ (r + 1)$ = The square lower 90 percent confidence level with $2\ (r + 1)$ degrees of freedom.

r = Number of chargeable failures.

4.7.1 Data collected will consist of all equipment malfunctions, number of hours completed when malfunction occurred, nature and cause of malfunction, time required to correct malfunction, and required corrective action.

4.7.2 Failure definition. A failure is any malfunction which renders the servo electronics unit in operative or operative outside the limits of this specification. In determination of chargeable failure, the following clarification shall apply.

4.7.2.1 If upon failure analysis (to detemine the cause of the malfunction) an assignable cause (related to manufacturing process, procedure, basic design weakness, etc.) is determined as the basis of the malfunction, the malfunction will not be chargeable as a reliability failure. However, the correction to the manufacturing process or procedure (et al) shall be accomplished and the testing resumed at the environment at which the malfunction occurred.

4.7.2.2 If upon failure analysis no assignable cause is evident and the malfunction is considered random in nature, the malfunction shall be a "chargeable" reliability failure and as such must be considered in the above demonstration test equation (4.7).

4.7.2.3 The decision as to chargeable reliability failures must be coordinated with the procuring activity (US Army Armament Research and Development Command, ATTN: DRDAR-QAF-A, Dover, NJ 07801.

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5. PREPARATION **FOR DELIVERY**

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10548193. The level of protection shall be as specified in the procurement document.

6. **NOTES**

6.1 Intended use. The two servo electronics are housed as part of the Power Supply, PP-4812/VPS-2. They amplify the input signals to produce the required power to drive the traverse and elevation motor of the Antenna Assembly, AS-2049/VPS-2.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, **number, and date of this specification.**

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. **Block of serial numbers to identify each servo electronics.**

g. Requirement for first article submission.

h. Availability of GFM Antenna Assembly AS-2049/VPS-2, **10556312** and Antenna Mount 10549103.

6.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring of the equipment. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Command, ATTN : DRDAR-QAF-I, Dover, NJ 07801.

When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of transmittal:

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a. The contract number.
b. The contract item (name, model number, etc.}.
The designs remaining to be submitted and the expected date of transmittal.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM .

6.5 Changes to previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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